

3DGS: 3D $\rightarrow$ 2D splatting math

* Define 3D Gaussian

$$G(\underline{x}) = e^{-\frac{1}{2}(\underline{x}-\underline{\mu})^T \underline{\Sigma}^{-1}(\underline{x}-\underline{\mu})}$$

$\underline{x}$: position in 3D space

$\underline{\mu}$: Center of the Gaussian

$\underline{\Sigma}$: 3x3 covariance matrix

$$\underline{\Sigma} = \underline{R} \underline{S} \underline{S}^T \underline{R}^T$$

$$\underline{S} = \begin{bmatrix} s_x & 0 & 0 \\ 0 & s_y & 0 \\ 0 & 0 & s_z \end{bmatrix}$$

learned Gaussian
params

$\underline{\mu}$, $\underline{\Sigma}$, $\underline{\alpha}$,

$\underline{c}$, $\underline{d}$,

$$\underline{S} = \exp(\underline{S}_{\text{raw}})$$

learned $\underline{S}_{\text{raw}}$ is pushed
high activation to ensure to

$\underline{z} \in \mathbb{R}^4$ | eared quaternion

$$\underline{z} = \frac{\underline{z}_{rw}}{\|\underline{z}_{rw}\|}$$

$$= \begin{bmatrix} w \\ x \\ y \\ z \end{bmatrix}$$

$$R = \begin{bmatrix} 1 - 2(y^2 + z^2) & 2(xy - wz) & 2(xz + wy) \\ 2(xy + wz) & 1 - 2(x^2 + z^2) & 2(yz - wx) \\ 2(xz - wy) & 2(yz + wx) & 1 - 2(x^2 + y^2) \end{bmatrix}$$

3D $\rightarrow$ 2D : Elliptical ^(EWA) Weighted Average
Volume (plating) (Cron 2001)

goal : 2D covariance matrix Σ'
from 3D " " Σ

* linear transformation M on
a Gaussian $\rightarrow$ covariance Σ will
result in Gaussian $\rightarrow$ covariance
 $M \Sigma M^T$

* but camera perspective projection
is non-linear (depth effect)

* affine approximation

* $\Sigma' = J W \Sigma W^T J^T$

W : viewing transformation matrix.
world coordinate $\rightarrow$ camera coordinate.

$W \Sigma W^T$ is the covariance
of 3D Gaussian in camera coordinate

J : Jacobian of the affine approx.
of the projective transformation.

W : camera extrinsics matrix.
 $\in \mathbb{R}^{3 \times 4}$

$$W = \begin{pmatrix} R_{\text{view}} & t_{\text{view}} \end{pmatrix}$$

which is camera
position

camera
location

J : ECA

Consider camera perspective projection

map 3D point (x, y, z) in
camera coordinate to 2D screen coordinate (u, v)

$$u = f_x \frac{x}{z} \quad \text{①}$$

$$v = f_y \frac{y}{z} \quad \text{②}$$

non linear $1/z$

first order Taylor approximation:

Calculate partial derivative of ① & ②
at the center of 3D gaussian
in camera coordinate $(t_x, t_y, t_z) = t$

$$\frac{\partial u}{\partial x} = \frac{f_x}{t_z}$$

$$\frac{\partial v}{\partial x} = 0$$

$$\frac{\partial u}{\partial y} = 0$$

$$\frac{\partial v}{\partial y} = \frac{f_y}{t_z}$$

$$\frac{\partial u}{\partial z} = -\frac{f_x t_x}{t_z^2}$$

$$\frac{\partial v}{\partial z} = -\frac{f_y t_y}{t_z^2}$$

$$J = \begin{bmatrix} f_x/t_z & 0 & -f_x t_x/t_z^2 \\ 0 & f_y/t_z & -f_y t_y/t_z^2 \\ 0 & 0 & 0 \end{bmatrix}$$

action
rectified flow
method